Reconstruction and the Sampling Theorem

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Key Concepts

1) The Fourier transform (FT) of a sampled signal $X_s(\Omega)$ is a sum of shifted replicates of the FT of the original continuous-time signal $X(\Omega)$. The replicates are shifted by integer multiples of the sampling frequency $\Omega_s = \frac{2\pi}{T}$.

$$X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(\Omega - k\Omega_s)$$

2) If $X(\Omega) = 0$ for $|\Omega| > B$ and $B < \Omega_s/2$ then the replicates of $X(\Omega)$ do not overlap.

a) The original FT $X(\Omega)$ is reconstructed from $X_s(\Omega)$ by low pass filtering to eliminate all replicates.

b) This corresponds to ideal sinc function interpolation in the time domain.

3) If $X(\Omega) = 0$ for $|\Omega| > B$ and $B > \Omega_s/2$ then the replicates of $X(\Omega)$ overlap in $X_s(\Omega)$.

a) The overlap results in aliasing - continuous-time frequency components appear at multiple frequencies in the discrete-time signal.

b) Cannot uniquely recover $x(t)$ from $x[n]$ due to the overlap.

4) The sampling theorem states the conditions under which there is no overlap in the shifted replicates for band limited signals.

   a) If $X(\Omega) = 0$ for $|\Omega| > B$, then $x(t)$ is uniquely determined by samples $x(nT)$ provided the sampling frequency $\Omega_s = \frac{2\pi}{T} > 2B$.

   b) $\Omega = B$ is called the Nyquist frequency.

   c) $\Omega_s = 2B$ is called the Nyquist sampling rate.

Video Screenshots

Reconstruction and The Sampling Theorem

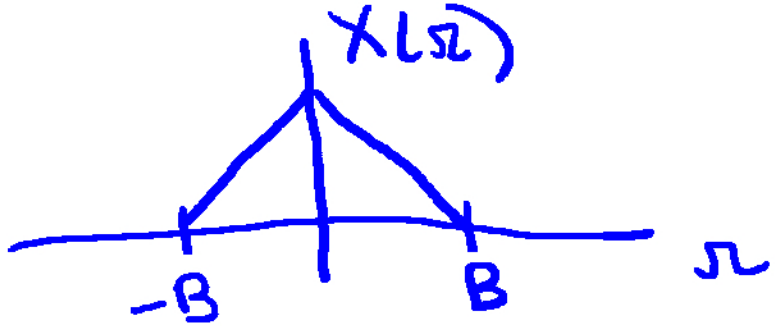
Recall - If $x(t) \xleftrightarrow{FT} X(\Omega)$ and $x[n] = x(nT)$, then

$$x[n] \xleftrightarrow{FT} X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(\Omega - k\Omega_s)$$

$\Omega_s = \frac{2\pi}{T}$: Sampling frequency $\left(\frac{1}{T} Hz\right)$

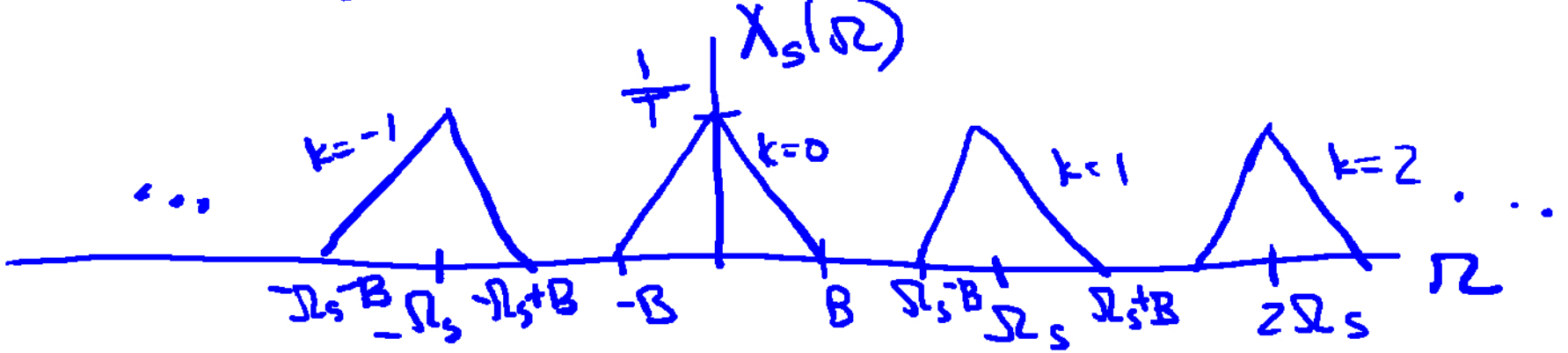
Reconstruction: find $x(t)$ given $x[n]$

Assume $x(t)$ is bandlimited

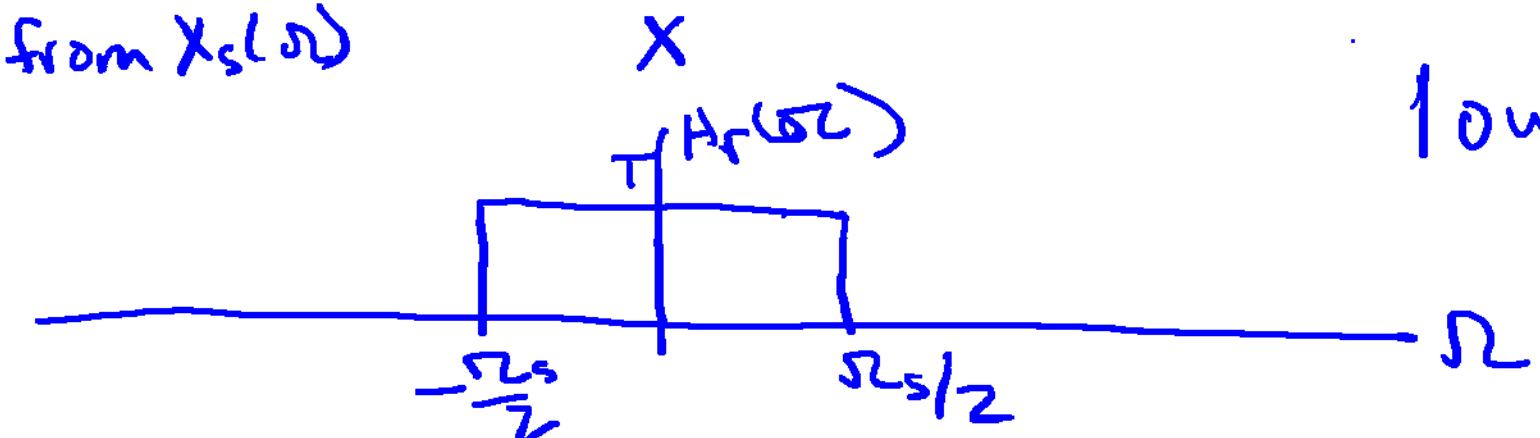


$X(\Omega) = 0$, $|\Omega| > B$

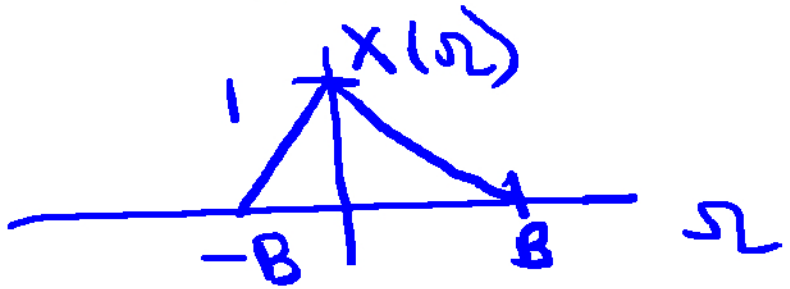
① Conside $\Omega_s > 2B$



Get $X(\Omega)$ from $X_s(\Omega)$



low pass filter

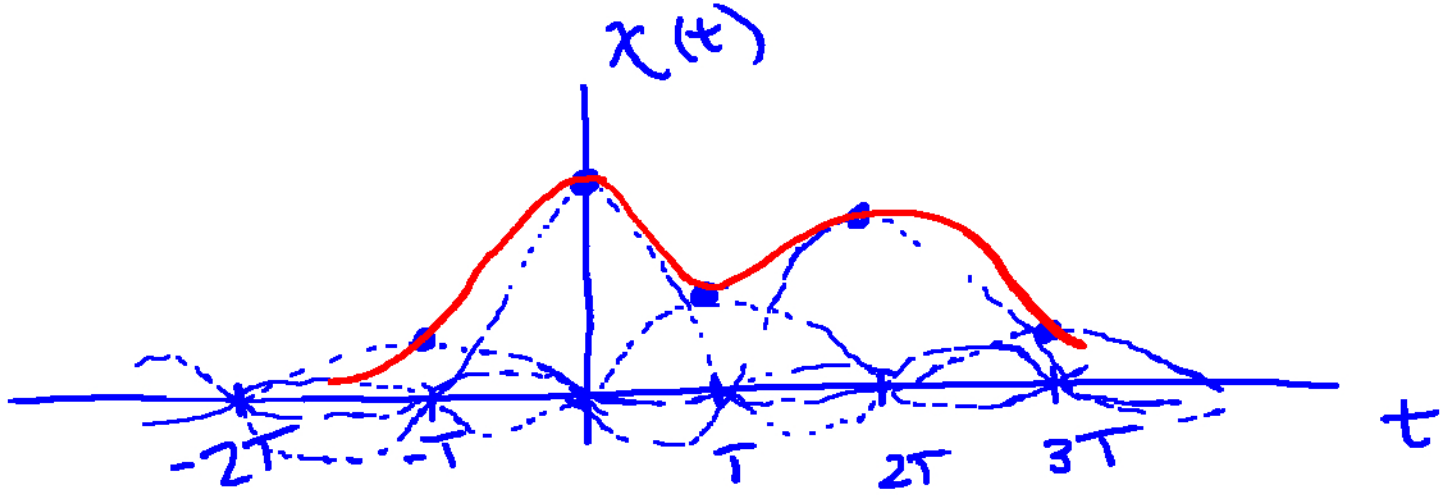


$$X(\Omega) = X_s(\Omega) H_r(\Omega) \xleftrightarrow{FT} x(t) = \sum_{n=-\infty}^{\infty} x[n]\delta(t-nT) * h_r(t)$$

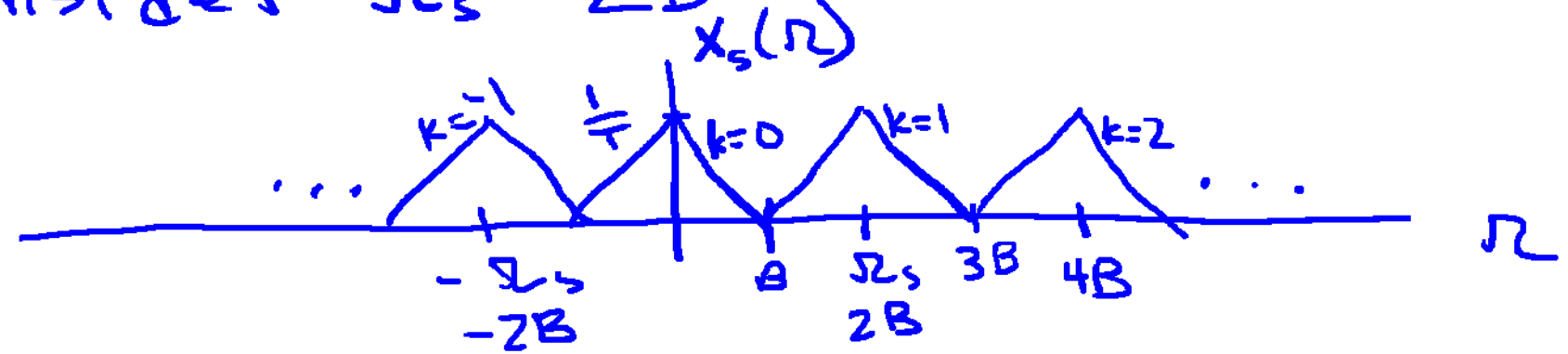
$$x(t) = \sum_{n=-\infty}^{\infty} x[n] h_r(t-nT) \qquad h_r(t) = T \frac{\sin\left(\frac{\Omega_s}{2} t\right)}{\pi t}$$

$$x(t) = T \sum_{n=-\infty}^{\infty} x[n] \frac{\sin\left(\frac{\Omega_s}{2}(t-nT)\right)}{\pi(t-nT)}$$

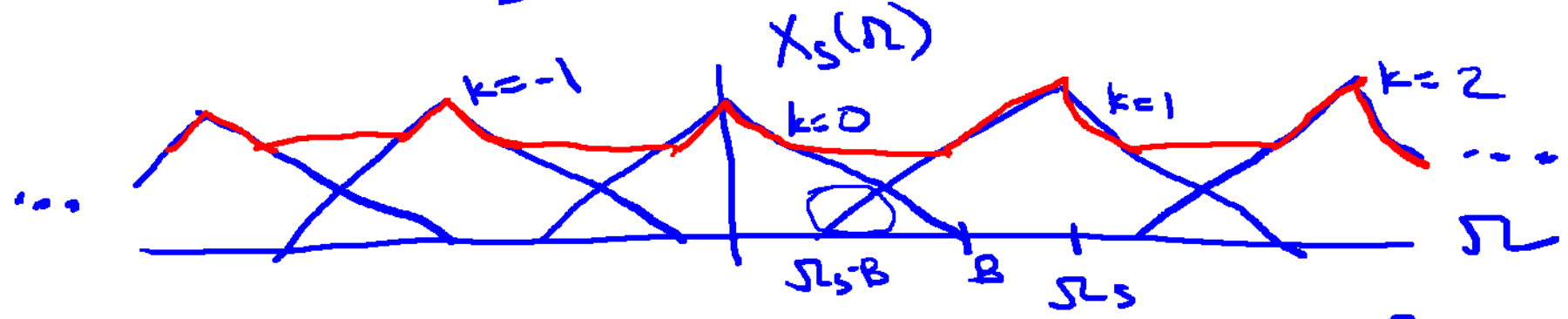
ideal bandlimited interpolation



(2) Consider $\Omega_s = 2B$



(3) Considers $\Omega_s < 2B$



overlap - cannot uniquely recover $x(t)$ from $x[n]$

Aliasing - frequency comps. in $X(\Omega - \Omega_s)$ show up in the "wrong" place

Sampling Theorem (Nyquist)

Let $x(t)$ be bandlimited with $X(\Omega)=0$ for $|\Omega|>B$

Then $x(t)$ is uniquely determined by its samples

$x[n]=x(nT)$ provided

$$\Omega_s=\frac{2\pi}{T}>2B$$

$\Omega=B$: Nyquist frequency

$\Omega=2B$: Nyquist rate

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